# Robust Pixel-by-Pixel Multimodal Remote Sensing Image Registration Using Geometry Preserving Dense Registration Network

Liang Zhou, Tao Peng, Zhiqiang Han, Liangzhi Li, Qing Zhu, and Yuanxin Ye, *Member, IEEE*

***Abstract*—The integration of multimodal remote sensing (RS) data is pivotal for advancing Earth observation capabilities, where high-precision cross-modal image registration serves as the cornerstone for realizing its full potential. While sparse feature matching methods have achieved high registration accuracy by extracting corresponding points and solving the parameters of the assumed geometric transformation model, they still struggle to effectively address local nonrigid deformations between multimodal images. Additionally, image dense registration methods are prone to introducing significant mismatching noise, which frequently results in structural distortions of typical ground features (e.g., roads, buildings, etc.) within global deformation regions. To address that, we propose a novel geometry-preserving dense registration network (GPDRNet) for multimodal RS image registration tasks. Our approach incorporates three key innovations: 1) a feature enhancement module that generates more discriminative cross-modal feature representations; 2) a geometry consistency loss that preserves critical structural and morphological characteristics of typical ground features within global deformation regions; and 3) a local deformation-aware postprocessing module that first estimates a global affine field, and then identifies local deformation regions through residual flow analysis. Experimental results validate the superior effectiveness and robustness of the proposed GPDRNet over existing state-of-the-art multimodal RS image registration methods. The code of the proposed method will be available at https://github.com/yeyuanxin110/GPDRNet.**

***Index Terms*—Dense registration, geometry preservation, global deformation, local nonrigid deformation, multimodal remote sensing (RS) image registration.**

## I. Introduction

AMIDST the rapid advancement of contemporary remote sensing (RS) technologies, the integrated application and analysis of multimodal RS data has emerged as a pivotal driving force in advancing related fields. The performance of RS image interpretation tasks, including semantic segmentation [1], [2], object detection [3], [4], and change detection [5], [6], can be substantially enhanced by fully leveraging the unique advantages of various modalities such as visible light, infrared, and radar. However, to fully unlock the potential of multimodal data, precise registration between multimodal RS images represents an indispensable prerequisite. This critical alignment process ensures effective information complementarity and synergistic integration across modalities, providing a solid foundation for further analysis and applications. Although multimodal RS image registration is of crucial importance, the substantial differences among different sensors pose significant challenges to this process [7]. A representative case is the co-registration of optical and synthetic aperture radar (SAR) imagery. Optical sensors operate on a passive imaging mechanism, capturing solar radiation reflected from the Earth's surface, which renders them vulnerable to atmospheric interference, particularly cloud obscuration and illumination variations. Conversely, SAR systems employ an active microwave imaging paradigm, providing superior all-weather and day-night detection capabilities. This fundamental divergence in imaging mechanism leads to significant nonlinear radiometric and geometric differences between optical and SAR data [8].

In response to these challenges, extensive research efforts, encompassing both conventional [9], [10], [11] and learning-based [12], [13], [14] approaches, have been dedicated to enhancing the accuracy of multimodal RS image registration. Most of these approaches initially identify sparse corresponding points between multimodal images and then solve for the unknown coefficients of an assumed geometric transformation model, such as affine or projective transformations [15], [16], to achieve image registration. While these methods can accurately model global deformations, they exhibit limited adaptability to local nonrigid deformations present in multimodal RS images. This issue is particularly evident in images acquired by SAR systems. Due to their side-looking imaging characteristics, SAR systems can cause local distortions such as foreshortening, layover, and shadow in areas with significant terrain undulations [17]. Under these circumstances, the geometric mapping between multimodal images becomes highly nonlinear, rendering it infeasible to accurately model with a single global geometric transformation function. In an effort to tackle local distortions, some studies have turned their attention to local parametric models [18], [19]. However, the difficulty of extracting sufficiently dense and uniformly distributed corresponding points between multimodal images has significantly limited the application of such models.

Received 17 September 2025; revised 25 December 2025 and 24 January 2026; accepted 9 February 2026. Date of publication 13 February 2026; date of current version 27 February 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFC3015404, in part by the National Natural Science Fund of China under Grant 42271446, and in part by the Science and Technology Program of Tianjin under Grant 24YFYSHZ00080. *(Corresponding author: Yuanxin Ye.)*

Liang Zhou, Tao Peng, Zhiqiang Han, Qing Zhu, and Yuanxin Ye are with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu 611756, China (e-mail: yeyuanxin@home.swjtu.edu.cn).

Liangzhi Li is with the School of Land Engineering, Chang'an University, Xi'an 710064, China (e-mail: liliangzhi@chd.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3664764

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

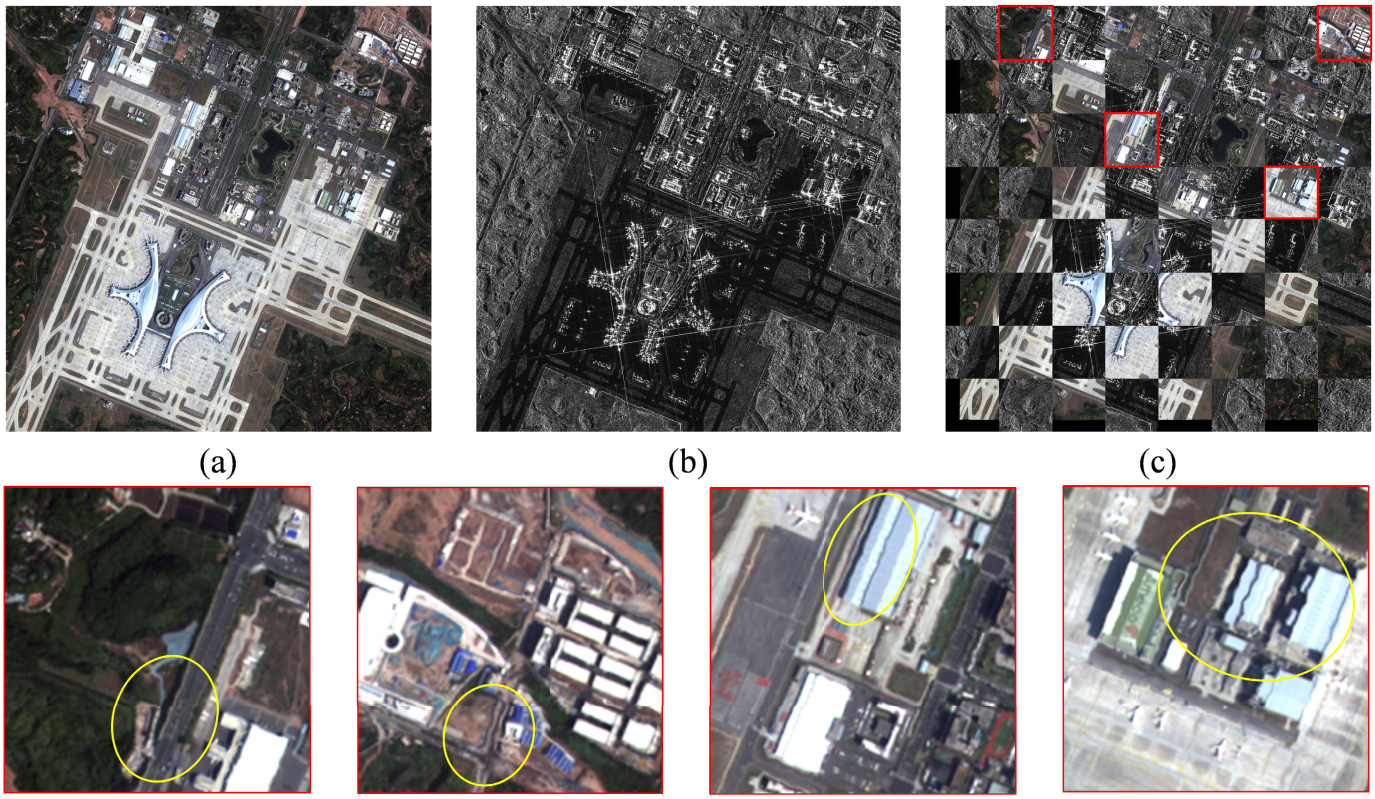



Fig. 1. Results of the multimodal RS image registration based on flow fields. (a) Moving image. (b) Fixed image. (c) Registration outcome depicted in a checkerboard pattern. Red rectangles highlight several areas where significant structural distortions of typical ground features are observed. The second row provides enlarged subimages from the red rectangle, with yellow ellipses further demarcating the areas of structural distortions.

Recently, dense registration methods [20], [21], [22] have gradually emerged as a solution for addressing RS image local nonrigid deformations. Unlike sparse matching, dense matching aims to establish the correspondence for each pixel in the image. Zhang et al. [20] developed a deep optical flow framework that achieves dense registration by modeling the optical flow field between multimodal RS images. Wang et al. [21] addressed local nonrigid deformations by effectively leveraging multiscale and multilevel image features to progressively refine flow field estimation. However, both methods exhibit notable limitations in handling rotation and scale variations, with their effectiveness confined to registration scenarios involving only minor geometric deformations. Xiao et al. [22] integrated affine and deformable registration modules to simultaneously address both global and local deformations in multimodal images. However, the reliance on conventional convolutional neural network (CNN) architectures for feature extraction constrained the model's feature representation capacity, particularly in comparison to modern Transformer-based pretrained models that have demonstrated superior performance. Additionally, while dense registration effectively handles local nonrigid deformations, the estimation of displacement vectors for each pixel inevitably introduces noise from false matches. Directly applying such deformation fields for image registration can easily result in structural distortions of typical ground features (e.g., roads, buildings, etc.) within global deformation regions, as shown in Fig. 1. Consequently, the performance of existing multimodal dense registration methods still requires further enhancement.

In this article, we propose a novel geometry-preserving dense registration network (GPDRNet) for multimodal RS image registration tasks, which can effectively address the challenges of global geometric deformation and local nonrigid deformation. The network adopts a coarse-to-fine registration framework. In the coarse registration stage, a pretrained Transformer network is employed as the encoder to extract coarse features, while in the fine registration stage, a CNN is utilized to capture fine feature representations. GPDRNet integrates a feature enhancement (FE) module, which employs attention mechanisms to enable cross-modal feature interaction and enhancement. Critically, GPDRNet incorporates a geometry consistency loss (GCL) during training to preserve the geometric integrity of typical ground features within global deformation regions. This is achieved by first partitioning the flow field into dense grids and then suppressing excessive deformation of grid cells and promoting the collinearity of continuous edge pairs across adjacent grids. Furthermore, we introduce a local deformation-aware postprocessing (LDAPP) module during the testing phase, which first estimates the global affine field and then identifies local deformation regions through residual flow analysis. By independently modeling both global and local deformations, GPDRNet achieves robust and highly accurate registration of multimodal RS images.

The main contributions of this study are as follows.

1) We integrate an FE module during the fine registration stage, which enhances the interaction of features from different modalities and thereby generates more discriminative feature representations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

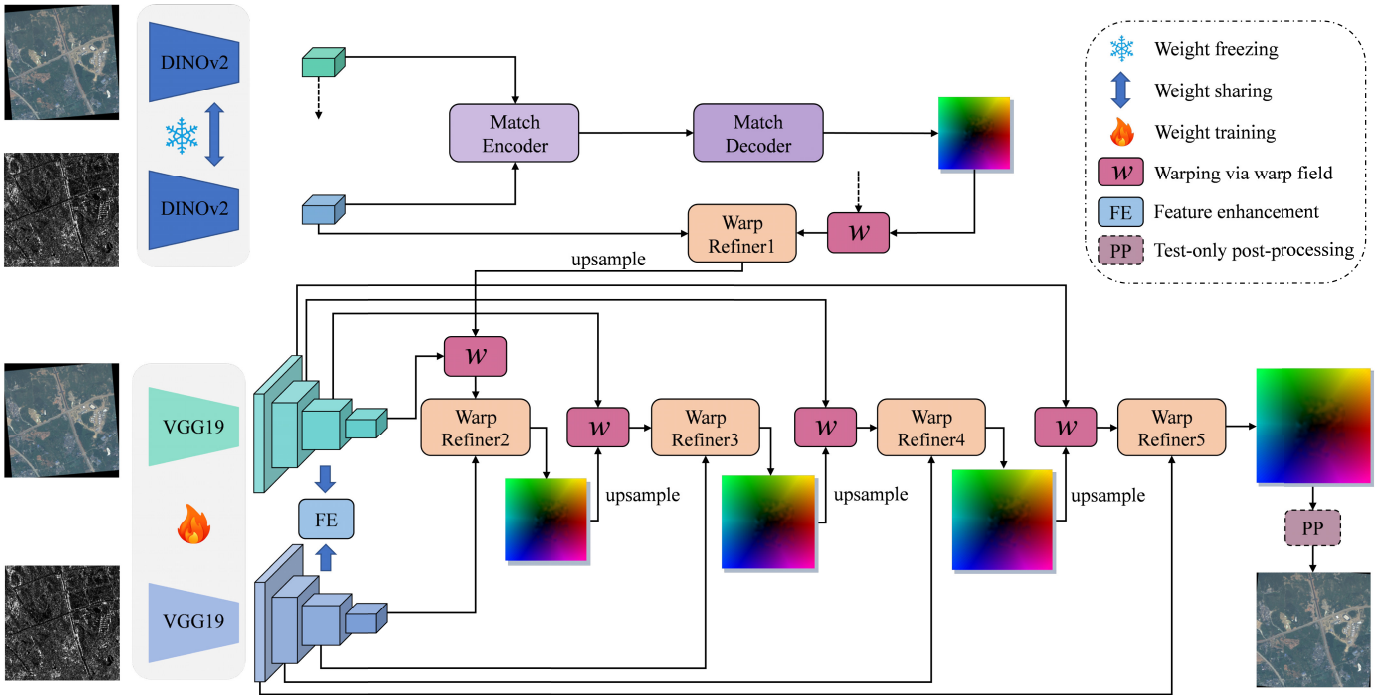



Fig. 2. Overview of our proposed GPDRNet.

2) We design a GCL to preserve the critical structural and morphological characteristics of typical ground features within global deformation regions.
3) We propose an LDAPP module during the testing phase, which separates local nonrigid deformation from global affine field to more effectively model the deformation field between multimodal images.

The subsequent sections of this work are organized as described below. Section II provides an overview of relevant works. The proposed GPDRNet is thoroughly detailed in Section III. Section IV conducts comparative experiments and presents a comprehensive discussion of the results. The article culminates in Section V with a summation of the principal contributions and proposals for subsequent research directions.

## II. Related Works

### A. Image Sparse Matching

In the early stages, nearly all image registration methods involve first identifying sparse corresponding points between images and then solving for the parameters of a predefined geometric transformation model to achieve registration. The focus of research is on improving the matching accuracy of these sparse points, which can be categorized into three main approaches: area-based, feature-based, and learning-based matching methods [23].

Area-based matching (i.e., template matching) methods search for correspondences by sliding a template from the sensed image over local regions from the reference image [24]. When the similarity measure selected reaches an extremum, the center point of the reference image region corresponding to this extremum and the center of the template are identified as a pair of matching points. However, conventional similarity measures, like the sum of squared differences (SSD) and the normalized cross correlation (NCC) [25], exhibit sensitivity to substantial nonlinear radiation differences (NRD). Designing robust feature descriptors to replace the original intensity information has become an effective strategy, giving rise to numerous related studies that encompass descriptors based on phase congruency [26], [27] and local self-similarity [9], [28]. Nevertheless, area-based methods are limited in their ability to handle scale and rotation differences between images, restricting their application scenarios.

Feature-based methods extract distinctive features from paired images and establish correspondences for matching, exhibiting enhanced robustness to geometric transformations. These methods typically detect key points in the images and extract their feature descriptors to achieve matching. Algorithms like scale invariance feature transform (SIFT) [29] and speeded-up robust features (SURFs) [30] can extract descriptors that are invariant to scale and rotation, thereby overcoming geometric changes to some extent. However, these SIFT-inspired methods excel at matching single-modality images but struggle with multimodality. Consequently, numerous advanced descriptors resilient to NRD have been developed. Notably, radiation-variation insensitive feature transform (RIFT) [31] utilizes phase congruency and a novel index map to achieve radiation-invariant feature matching, enabling robust matching across multimodal images.

With artificial intelligence advancing, deep learning-based approaches have demonstrated remarkable potential in addressing image matching challenges [32]. For instance, Merkle et al. [33] developed a Siamese convolutional network incorporating dilated convolutions to enhance feature extraction, while Zhang et al. [34] proposed an advanced pseudo-Siamese framework with multilevel feature fusion and attention mechanisms to improve matching accuracy. Hughes et al. [35] achieved SAR-optical matching using goodness, cor-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

respondence, and outlier networks. Although these learning-based template matching methods employ deep feature representation to overcome NRD, they remain vulnerable to scale and rotation variations. In contrast, learning-based feature matching has garnered increased attention. The classic combination of SuperPoint [36], a self-supervised feature extractor, with Superglue [37], an attention-based matcher, forms a powerful duo that substantially boosts matching accuracy. Expanding on the Superglue framework, Lindenberger et al. [38] introduced LightGlue, an advanced feature matching approach that integrates novel architectural optimizations and an adaptive computation strategy. Based on the describe-then-detect paradigm, Chen et al. [39] constructed robust quadratic net (RQ-Net) with a dual-encoder–decoder structure, where one encoder captures local features, and the other establishes global relationships across and within modalities.

However, sparse matching methods exhibit significant limitations when dealing with the registration of multimodal images under complex mapping scenarios. When the mapping relationship between images cannot be modeled by a unified geometric transformation, the registration performance of sparse matching methods often deteriorates, failing to meet the demands of high-precision registration.

### *B. Image Dense Matching*

Dense matching aims to establish a dense pixel-wise correspondence field across images. Unlike sparse matching methods, dense matching captures the full spatial transformation between images, making it particularly suitable for handling complex nonrigid image deformations.

A widely employed dense matching technique is optical flow estimation, which computes a per-pixel displacement vector by optimizing an energy function that balances image similarity and spatial smoothness. Derived from the well-established Lucas–Kanade (LK) [40] method, eFolki [41] offers a computationally efficient approach for optical flow estimation specifically designed to achieve robust co-registration of urban SAR images under significant decorrelation. Further extending this, Brigot et al. [42] developed GeFolki, which combines a multiscale strategy and specific filtering algorithms to achieve dense registration among heterogeneous optical, light detection and ranging (LiDAR), and SAR sensors in forestry applications. To address the limitation of the brightness constancy assumption when applied to cross-modal scenarios, OS-Flow [43] introduces the dense structural feature descriptor gradient location orientation histogram (GLOH) to replace the original intensity information. The method formulates global and local optical flow estimation approaches based on the Horn–Schunck (HS) [44] and LK frameworks, while employing a coarse-to-fine strategy to handle large displacements. Similarly, Yu et al. [45] employed SIFT and phase congruency-based descriptors to mitigate radiometric differences between optical and SAR images, and further refined the optical flow estimation using an attention mechanism.

Despite these advances, traditional dense registration methods often rely on handcrafted features and iterative optimization, which can be sensitive to initialization and struggle with the lack of texture. In contrast, deep learning-based dense registration aims to learn optimal feature representations from training datasets, offering greater robustness to complex mapping. FloU-Net [46] leverages a U-Net [47] backbone to directly estimate optical flow fields for registering Sentinel-2 and Sentinel-3 images. This approach uniquely exploits the inherent resolution difference as a self-supervised signal, bypassing the need for ground-truth optical flow fields. Inspired by the advanced optical flow estimation method RAFT [48], Zhang et al. [20] introduced OSFlowNet, a robust deep optical flow framework specifically designed for dense registration of high-resolution optical and SAR images. OSFlowNet features a pseudo-Siamese network incorporating dilated feature concatenation to enhance feature representation, and employs a gated recurrent unit (GRU) module to perform the iterative optimization process of optical flow fields. Unlike direct flow estimation in FloU-Net or iterative refinement in OSFlowNet, Wang et al. [21] adopt a coarse-to-fine strategy that couples a ResNet [49] feature pyramid with a Swin-Transformer [50] dense matching module to achieve multihierarchy flow field cumulative prediction. Xiao et al. [22] propose affine and deformable registration networks (ADRNets), a hybrid framework unifying affine and deformable registration in a cascaded pipeline for multimodal RS imagery. ADRNet first regresses affine parameters to correct large-scale geometric deformations, and then refines the displacement field based on the U-Net equipped with dilated convolutional attention blocks.

Recently, learning-based dense feature matching approaches have leveraged robust feature representations to establish dense correspondences across images. Edstedt et al. [51] presented dense kernelized feature matching (DKM), the first dense matching framework to decisively outperform sparse methods in geometry estimation. DKM utilizes a Gaussian Process (GP) to better handle the uncertainty of matching, and refines the warp with large depth-wise convolutions on stacked feature maps. Building on DKM, robust dense feature matching (RoMa) [52] replaces the coarse encoder with a frozen Transformer-based backbone to extract robust features, and introduces a CNN-based fine encoder to deliver precise spatial localization during refinement. In addition, RoMa constructs a coarse matcher that discretizes the output space into anchor probabilities to better model multipeaked distributions. Ren et al. [53] proposed modality invariant image matching (MINIMA), a framework that enhances the generalization of existing dense matching methods through data expansion. This approach leverages a data engine to synthesize large-scale cross-modal image pairs. By fine-tuning models on this synthetic dataset, MINIMA achieves cross-modal invariance without architectural modifications, significantly boosting registration performance.

While dense matching generates abundant correspondences, they inevitably introduce substantial mismatch noise. Directly applying such deformation fields for image registration can easily result in structural distortions of typical ground features within global deformation regions. Therefore, developing a robust dense registration method to deal with complex nonrigid deformations and maintain the geometric integrity of ground features within global deformation regions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

remains a key challenge for accurate multimodal RS image registration.

## III. Methodology

Our method extends the dense registration architecture RoMa by constructing a novel FE module, a multisupervised loss, and a postprocessing module, significantly improving the effectiveness in handling complex global and local nonrigid deformations across multimodal images. In this section, we will systematically elaborate on the proposed GPDRNet. First, we provide a brief overview of the overall framework of our method. Subsequently, we delve into the pertinent details regarding the coarse and fine registration processes for multimodal RS images. Next, we introduce the multiple loss functions employed and detail the design and implementation of the proposed GCL. Finally, we discuss the module for identifying local deformation regions based on residual flow analysis.

### A. Framework Overview

The overall architecture of our proposed GPDRNet is illustrated in Fig. 2. The registration pipeline follows a coarse-to-fine strategy. In the coarse registration stage, a weight-sharing pretrained DINOv2 [54] network extracts the down-sampling deep features of the moving image and the fixed image. These features are subsequently processed through a match encoder and a match decoder to estimate the initial warp field and its corresponding confidence. During the fine registration phase, hierarchical multiscale features from VGG19 [55] are enhanced through an FE module to facilitate cross-modal information interaction, generating more discriminative feature representations. These enhanced features are subsequently integrated with the downsampled deep features extracted by DINOv2 to construct a feature pyramid. The warp field and its confidence are progressively refined at each feature scale until reaching full resolution, resulting in the final refined warp field between the multimodal image pairs. During testing, the final warp field undergoes further optimization via a postprocessing module. Finally, the moving image is warped based on the output deformation field to generate the registered result image.

### B. Coarse Registration

Following RoMa, we employ a GP as the matching encoder to generate matching-specific feature representations based on image coordinate embeddings. As a nonparametric approach, the GP inherently adapts to complex mapping relationships between images. Furthermore, it natively provides uncertainty quantification, which is beneficial for assessing the reliability of the matching results.

The original DINOv2 features are then concatenated with the features output by the GP to form a fused feature representation, which is fed into the Transformer match decoder. This decoder comprises five stacked vision transformer (ViT) blocks, each equipped with 8 attention heads, a hidden dimension of 1024, and a multilayer perceptron (MLP) dimension of 4096. Unlike the paradigm of directly regressing continuous coordinates, it predicts a discrete probability distribution over $K = 64 \times 64$ spatially uniform anchors $A_{k_{k=1}}^{K}$ covering the entire image plane, along with pixel-wise matchability.

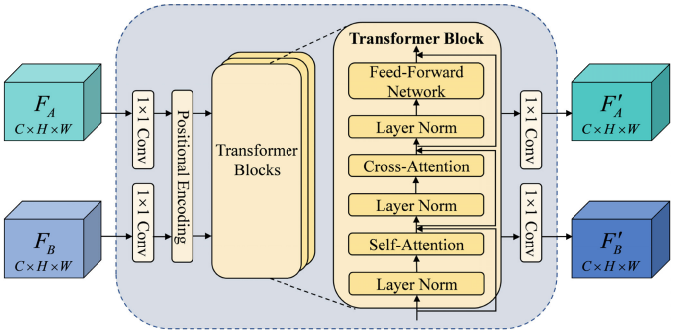



Fig. 3. Architecture of the FE module.

By selecting the optimal anchor via the argmax operation and performing a probability-weighted average with its four neighboring anchors, a smooth and continuous warp estimation can be obtained. This matching strategy combines the discreteness of classification with the subpixel precision of locally weighted regression, making it particularly suitable for handling multipeaked correspondence ambiguities, where a single point corresponds to multiple candidate match locations.

### C. Fine Registration

The refinement process is crucial for achieving high-precision matching, with the core objective of optimizing the coarse warp field and confidence obtained from the coarse registration stage. Compared to DINOv2 features rich in high-level semantics, the fine features extracted by VGG19 exhibit superior spatial localization capabilities but suffer from limited representational capacity. To enhance the discriminability of VGG19 features, we design a Transformer-based FE module. As shown in Fig. 3, it first uses a $1 \times 1$ convolution to project the input features to a hidden dimension of 128. These features are then processed by five stacked Transformer blocks. Each block alternately employs self-attention and cross-attention mechanisms and contains an MLP with a dimension of 512. Finally, the features are projected back to their original dimension using another $1 \times 1$ convolution. The self-attention mechanism captures intra-feature interactions within individual feature maps, enriching contextual representations. Conversely, cross-attention facilitates interfeature interactions between distinct feature maps, strengthening cross-modal dependencies. To reduce the computational cost, we employ shifted window attention [50], confining operations to local windows while maintaining global connectivity through cross-window connections. These enhanced feature maps exhibit greater discriminability and richer contextual information, providing more effective inputs for warp field optimization.

Subsequently, a multiscale feature pyramid is constructed by combining multilevel VGG19 features with DINOv2 downsampled features. The refinement process is an iterative procedure that begins with coarser scales and progressively refines to finer scales. At each scale, a specific convolutional refiner predicts relative offsets to progressively update both the warp field and confidence map. Notably, the refiner's input is adjusted across scales. For the three coarsest levels ($14\times$, $8\times$, $4\times$), it incorporates image features, displacement embeddings, and local correlation volumes. At the finest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

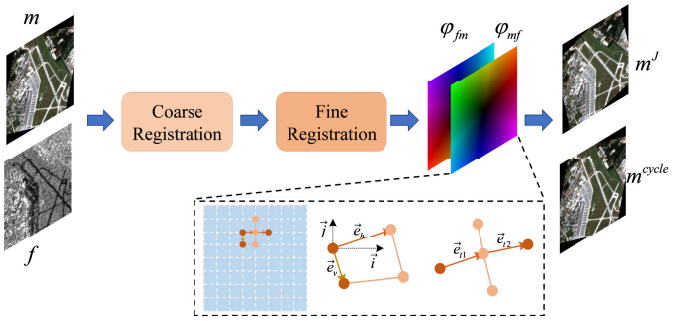



Fig. 4. Process of bidirectional registration and the construction of GCL.

two scales (2×, 1×), with the warp field and confidence sufficiently optimized, we omit correlation volumes to improve computational efficiency, preserving only image features and displacement embeddings for final refinement. Through this multistage propagation, the model converges to high-precision warp and confidence estimates at the original resolution.

### D. Multisupervised Loss

We perform bidirectional registration by simultaneously estimating both the warp field $\varphi_{mf}$ from the moving image $m$ to the fixed image $f$ and its inverse $\varphi_{fm}$ from the fixed image back to the moving image. Fig. 4 shows the registration process and the correspondence between the image and the abbreviation, where the moving image is first warped using $\varphi_{mf}$ to produce $m^J$, which is then reconstructed based on $\varphi_{fm}$ to obtain $m^{\text{cycle}}$.

During the coarse registration stage, for each point $x^s$ in the source image, the model predicts a probability distribution $p_k(x^s)$ over 64 × 64 anchors, along with a corresponding matchability $c(x^s)$. To optimize the predicted anchor probability, we employ a cross-entropy loss $L_{\text{CE}}$. Additionally, we employ a binary cross-entropy loss $L_{\text{BCE}}$ to evaluate the matchability. The total coarse loss combines both terms with a weighting factor $\lambda = 0.01$, which is defined as

$$L_{\text{coarse}} = L_{\text{CE}}\left(p_k\left(x^s\right), id\left(x^t\right)\right) + \lambda L_{\text{BCE}}\left(c\left(x^s\right), c_{gt}\left(x^s\right)\right) \tag{1}$$

where $id(x^t)$ denotes the index of the anchor closest to the ground-truth match point $x^t$. For each pixel in the source image, its ground-truth matchability is defined as

$$c_{gt}\left(x^s\right) = \begin{cases} 1, & \text{if } \left\|\mu\left(x^s\right) - x^t\right\| \le \tau \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\mu$ represents the coordinate predicted by the model, $\|\cdot\|$ denotes the L2 norm, and $\tau$ is set to 2 pixels in this work. In other words, a pixel is considered matchable if the deviation between its estimated location and the true location is within the tolerable range.

During the refinement process, the model predicts a residual offset at each scale $i$ to iteratively adjust the warp estimate from the previous coarser level. To handle outliers and ensure robustness during refinement, we adopt a generalized Charbonnier function for regression, which provides a continuous transition between L2-like behavior for small errors and more

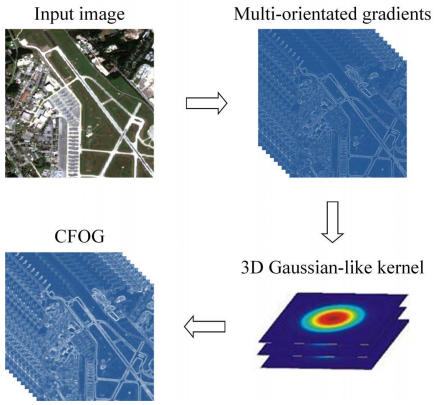



Fig. 5. Construction process of CFOG.

robust characteristics for large errors. The formula is as follows:

$$L_{\text{reg}}^i = \left(\left\|\mu\left(x^s\right) - x^t\right\|^2 + \beta_i^2\right)^{\alpha/2} \tag{3}$$

where $\beta_i = d \cdot 2^i$ with $d = 0.03$ is a scale-dependent parameter that adapts the loss sensitivity according to the current refinement level, and $\alpha = 0.5$ controls the robustness properties of the loss function. Similar to the coarse stage, we maintain a matchability evaluation throughout the refinement process. The total fine loss is thus defined as

$$L_{\text{fine}} = \sum_i L_{\text{reg}}^i + \lambda L_{BCE}\left(c^i\left(x^s\right), c_{gt}\left(x^s\right)\right). \tag{4}$$

To effectively constrain both global and local nonrigid deformations, we introduce additional supervisory signals at the original resolution, constructing a multisupervised loss to improve geometric consistency and structural fidelity of the warp field. To enhance the reversibility of the final warped field, we compute the reconstruction loss $L_{\text{recon}}$ after warping the moving image twice using the deformation fields obtained from bidirectional registration. The calculation is as follows:

$$L_{\text{recon}} = \left\|m - m^{\text{cycle}}\right\|. \tag{5}$$

To evaluate the registration performance between the moving and fixed images, we introduce a structural similarity loss. The channel features of oriented gradients (CFOGs) [24], as shown in Fig. 5, are used to compute the structural similarity loss between multimodal images due to their robustness and efficiency in capturing structural features. The specific computation process is as follows:

$$L_{\text{ssim}} = \text{SSD}\left(\text{CFOG}\left(f\right), \text{CFOG}\left(m^J\right)\right). \tag{6}$$

While dense matching establishes abundant correspondences, it simultaneously introduces significant mismatching noise, which easily results in structural distortions of typical ground features within global deformation regions. Drawing on the concept of image stitching [56], we construct a GCL $L_{\text{GCL}}$ to encourage the network to retain the geometric attributes of typical features. The deformation field is divided into $M \times N$ equally sized dense grids. Based on the warp field within each grid, the directions of its four vertices are regressed to derive the vectors $\vec{e}$ representing the grid's four edges. As shown in Fig. 4, by calculating $L_{intra}$ to prevent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

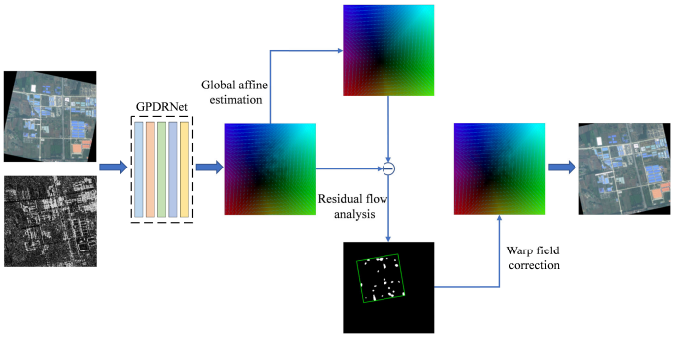



Fig. 6. Process of local deformation region identification.

excessive deformation of the grid shape and calculating $L_{inter}$ to encourage collinearity of adjacent edges in neighboring grids in horizontal or vertical directions, we form the geometry consistency constraint. The specific formulas are as follows:

$$L_{intra} = \frac{1}{(M+1)\times N}\sum_{\{\vec{e}_h\}}\sigma\left(\left\langle\vec{e},\vec{i}\right\rangle - \frac{2W}{N}\right) + \frac{1}{M\times(N+1)}\sum_{\{\vec{e}_v\}}\sigma\left(\left\langle\vec{e},\vec{j}\right\rangle - \frac{2H}{M}\right) \tag{7}$$

$$L_{inter} = \frac{1}{Q}\sum_{\{\vec{e}_{t1},\vec{e}_{t2}\}} 1 - \frac{\langle\vec{e}_{t1},\vec{e}_{t2}\rangle}{\left\|\vec{e}_{t1}\right\|\cdot\left\|\vec{e}_{t2}\right\|} \tag{8}$$

$$L_{GCL} = L_{intra} + L_{inter} \tag{9}$$

where $\{\vec{e}_h\}$, $\{\vec{e}_v\}$ represent the set of horizontal and vertical edges, $\vec{i}$, $\vec{j}$ are the horizontal and vertical unit vector, $\sigma(\cdot)$ represents the rectified linear unit (ReLU) function, and $Q$ is the total number of consecutive edge pairs. To balance the effect of geometry consistency constraints with computational efficiency, both $M$ and $N$ are uniformly set to 8.

Finally, we aggregate these loss terms to obtain the final loss. This combination of multiple loss functions provides the model with more comprehensive supervisory signals, thereby enhancing the model's robustness and generalization capability. The corresponding formula is presented below

$$L_{total} = L_{coarse} + L_{fine} + L_{recon} + L_{ssim} + L_{GCL}. \tag{10}$$

### E. Local Deformation Region Identification

To further mitigate structural distortions of typical features in global deformation regions and improve registration performance, we design an LDAPP module during the testing phase to separate the global and local deformation regions. As shown in Fig. 6, after obtaining the final warp field and its confidence map, we first apply a threshold of 0.7 to retain only high-confidence pixels. From these candidate pixels, we perform grid-based sampling to ensure spatial uniformity. The image is divided into $8\times 8$ nonoverlapping cells. Within each cell, the top 10% of pixels with the highest confidence are selected. This strategy ensures that the sampled points are both reliable and distributed across the entire image. The optimal affine transformation matrix is then estimated using the random sample consensus (RANSAC) algorithm, maximizing the alignment with the global motion trend in the warp field. Based on the estimated global affine transformation, the residual flow is computed by subtracting the affine field from the original warp field.

The residual field captures the remaining local deformations after global motion compensation. The magnitude of the residual flow field is first binarized using Otsu's thresholding method, generating an initial distortion mask. Directional coherence is then assessed by computing the variance of flow angles within local windows. The final distortion mask is determined by combining Z-score normalized magnitudes and directional consistency results. To refine the mask, morphological opening is applied to suppress isolated noise, followed by connected-component analysis using a minimum-area threshold to eliminate small spurious regions.

Following the identification of local deformation regions, the warp field within the global deformation area is adjusted to conform to an affine transformation model. The corrected warp field is subsequently applied to align the moving image with the fixed image. This postprocessing strategy can effectively enhance the registration performance of multimodal RS images.

## IV. Experiments and Results

In this section, we begin by presenting the publicly available multimodal RS datasets used in our experiments. We then describe the implementation details of our method, along with the evaluation metrics for assessing registration performance. In addition, comparative tests against existing state-of-the-art image registration algorithms are reported from both quantitative and qualitative perspectives. Furthermore, we perform ablation experiments to validate the contribution of each key module in our proposed GPDRNet, highlighting their individual roles in improving registration accuracy.

### A. Datasets

To ensure experimental reproducibility, we employ two publicly available optical-SAR datasets: OSdataset [57] and WHU-OPT-SAR [58]. OSdataset comprises 2673 co-registered optical-SAR patch pairs with a size of $512\times 512$ pixels and a resolution of 1 m. SAR data are acquired by the GaoFen-3 satellite's spotlight mode, while optical images are sourced from Google Earth, covering 20 global urban and rural locations. The dataset has already been officially split into training, validation, and test subsets. Accordingly, we follow the provided split of 2011, 238, and 424 image pairs, respectively. WHU-OPT-SAR features 100 co-registered image pairs spanning roughly 50 000 $km^2$ across Hubei Province. All these images are partitioned into nonoverlapping patches sized $512\times 512$ pixels, yielding 7000 SAR–optical pairs that are subsequently partitioned into training, validation, and test subsets at a 6:2:2 ratio.

To address complex registration scenarios that involve both global transformations and local nonrigid deformations, we synthesize samples based on controlled random warping. Specifically, global deformations are implemented through affine transformations parameterized with scaling factors ranging from [0.8, 1.2], translations within [−15, 15] pixels, and rotations limited to [−20, 20]. Nonrigid deformations are introduced via elastic transformations. We follow a common

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

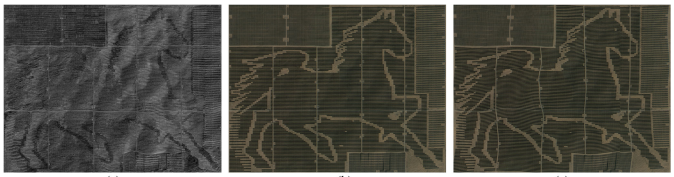


Fig. 7. Validation of synthetic nonrigid deformations. (a) SAR image with natural nonrigid distortion. (b) Corresponding optical image. (c) Optical image after elastic transformation.

TABLE I
COMPARATIVE RESULTS OF ALL METHODS ON THE OSDATASET

| Methods | Metrics | | | |
|---|---|---|---|---|
| | $CMR_{T\leq 1px}$ | $CMR_{T\leq 3px}$ | $CMR_{T\leq 5px}$ | *EPE* |
| OS-Flow | 13.14 | 25.82 | 38.64 | 9.24 |
| RIFT | 19.84 | 52.74 | 75.29 | 4.52 |
| OSFlowNet | 17.66 | 57.97 | 83.05 | 3.93 |
| ADRNet | 19.53 | 61.10 | 84.81 | 3.59 |
| RoMa | 22.42 | 69.27 | 87.86 | 3.03 |
| $\text{MINIMA}_{\text{RoMa}}$ | 25.16 | 73.56 | 90.46 | 2.44 |
| Ours | **25.98** | **74.85** | **92.33** | **2.26** |

practice of simulating corresponding deformations on optical data for clearer visual assessment, although local nonrigid deformations are prominent in SAR imaging. To evaluate the realism of these synthetic deformations, we conduct a visual comparison between a 0.35-m resolution Capella SAR image captured from Inner Mongolia, China, and its corresponding optical counterpart. As illustrated in Fig. 7, the SAR image exhibits pronounced nonrigid distortions, while the optical image remains relatively rigid. By applying an elastic transformation to the optical image, we observe that the synthetic deformation effectively approximates the real nonrigid distortion present in the SAR image.

We employ distinct transformation strategies for training and testing phases. During training, we utilize two transformation pipelines: 1) random affine-only transformations and 2) random affine transformations followed by elastic transformations. For testing, synthesized images undergo global affine transformations combined with local elastic deformations, where local elastic transformations are constrained to [1/4, 1/3] of the total image area.

### B. Implementation Details and Evaluation Metrics

The implementation of the proposed method is carried out using the PyTorch framework, initialized with pretrained weights. For the training process, the AdamW optimizer is utilized, with an initial learning rate of 0.001 and a maximum of 200 training epochs. Our experiments are performed on a server with dual Intel Xeon Platinum 8468V CPUs and four NVIDIA GeForce RTX 4090 GPUs. During training, the batch size is set to 4, and the image pairs are resized to $280 \times 280$ pixels for model input.

We select correct matching rate (*CMR*) and endpoint error (*EPE*) as evaluation metrics. *CMR* quantifies registration accuracy by calculating the proportion of correspondences where the L2 norm between the estimated warp field and ground truth falls within specified thresholds (e.g., 1, 3, or 5 pixels in this work). *EPE* calculates the average Euclidean distance between the estimated and actual warp fields.

TABLE II
COMPARATIVE RESULTS OF ALL METHODS ON THE WHU-OPT-SAR

| Method | Metrics | | | |
|---|---|---|---|---|
| | $CMR_{T\leq 1px}$ | $CMR_{T\leq 3px}$ | $CMR_{T\leq 5px}$ | *EPE* |
| OS-Flow | 11.73 | 20.84 | 36.67 | 10.36 |
| RIFT | 18.20 | 49.45 | 73.35 | 5.72 |
| OSFlowNet | 16.99 | 57.43 | 80.41 | 4.15 |
| ADRNet | 20.63 | 60.56 | 83.25 | 3.84 |
| RoMa | 23.28 | 71.52 | 87.90 | 2.87 |
| $\text{MINIMA}_{\text{RoMa}}$ | 26.93 | 76.07 | 91.68 | 2.35 |
| Ours | **27.55** | **77.59** | **92.61** | **2.18** |

### C. Registration Result Analysis

This section presents a comparative analysis between our proposed GPDRNet and existing state-of-the-art image registration methods, including RIFT based on sparse feature matching, along with OS-Flow, OSFlowNet, ADRNet, RoMa, and $\text{MINIMA}_{\text{RoMa}}$ as representative dense registration approaches.

RIFT [31], which achieves radiometric-invariant multimodal matching by combining phase-consistency-based maximum index maps with sparse feature point matching and outlier rejection to handle global deformations.

OS-Flow [43], which replaces intensity information with the GLOH descriptor and estimates dense optical flow between multimodal images based on conventional HS and LK frameworks.

OSFlowNet [20], which extracts discriminative features using a pseudo-Siamese network incorporating dilated concatenation, and employs a GRU module that iteratively updates and optimizes the dense optical flow fields.

ADRNet [22], which first employs an affine registration module to regress six affine parameters, followed by a U-Net-based deformable registration module to refine the residual displacement field at the pixel level.

RoMa [52], which integrates frozen coarse features with specialized fine features, employs a Transformer decoder predicting anchor probabilities, and designs a robust loss for dense matching.

$\text{MINIMA}_{\text{RoMa}}$ [53], which leverages a data engine to synthesize a large-scale cross-modal image dataset for fine-tuning RoMa, enhancing dense matching generalization and registration performance without architectural modifications.

*1) Quantitative Analysis:* We quantitatively evaluate the registration performance of each method using the *CMR* and *EPE* metrics, with detailed results presented in Tables I and II. It can be observed that OS-Flow demonstrates relatively poor registration performance on both OSdataset and WHU-OPT-SAR datasets, as evidenced by its lower *CMR* and higher *EPE* values. This performance limitation of OS-Flow stems primarily from its reliance on the handcrafted GLOH descriptor to maintain the constant brightness assumption in optical flow

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

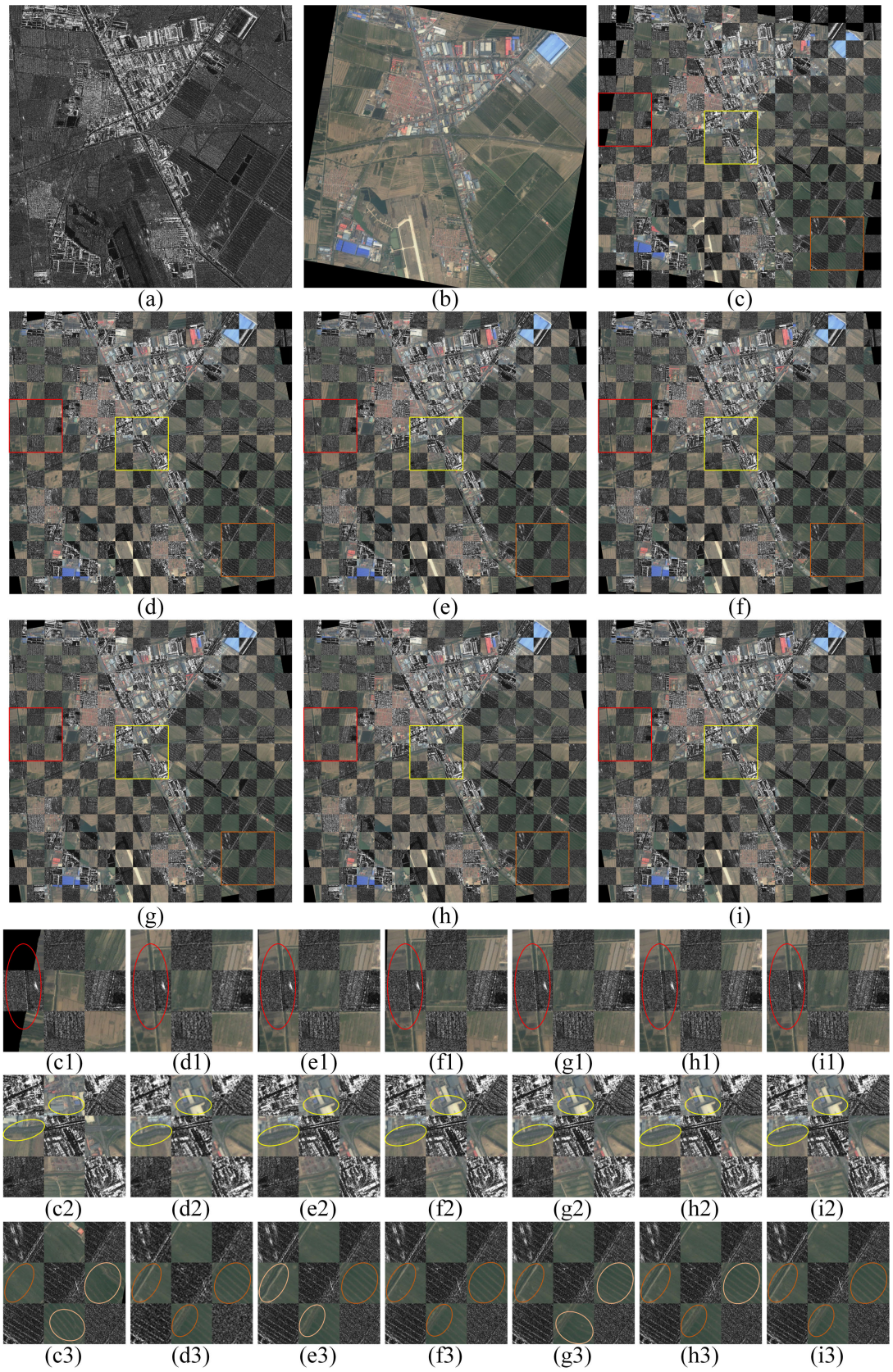



Fig. 8. Mosaic images of the representative registration result. (a) Fixed image. (b) Moving image. (c) OS-Flow. (d) RIFT. (e) OSFlowNet. (f) ADRNet. (g) RoMa. (h) $\text{MINIMA}_{\text{RoMa}}$. (i) GPDRNet. The red rectangles indicate the regions of global affine deformation, the yellow rectangles highlight the areas of local nonrigid deformation, and the orange rectangles mark the locations within the global deformation regions where the geometric structures of typical ground features are prone to change. (c1)–(i1), (c2)–(i2), and (c3)–(i3) are enlarged subimages from the red, yellow, and orange rectangles of (c)–(i), respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

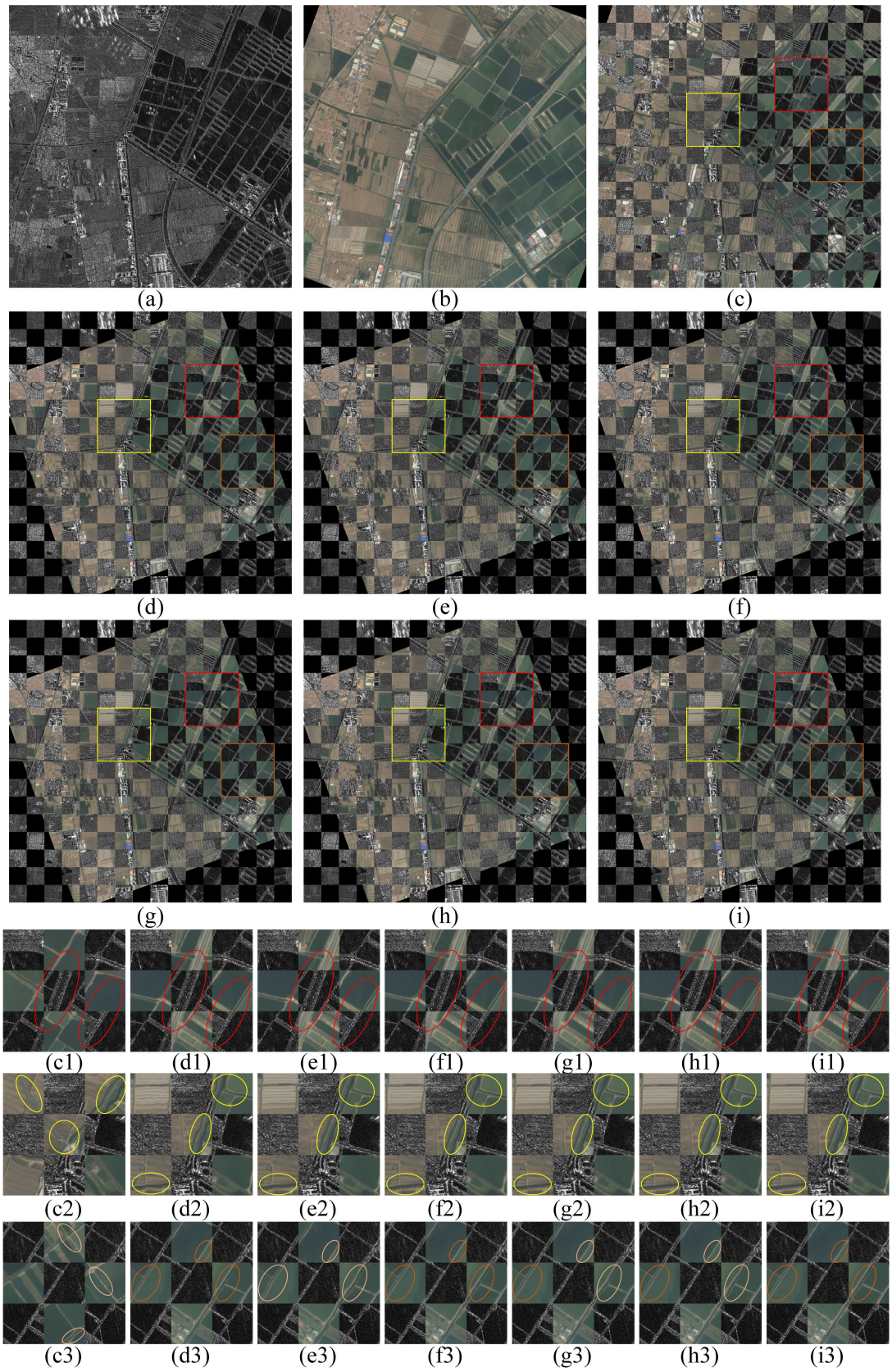



Fig. 9. Mosaic images of the registration result under large deformation magnitudes. (a) Fixed image. (b) Moving image. (c) OS-Flow. (d) RIFT. (e) OSFlowNet. (f) ADRNet. (g) RoMa. (h) $\text{MINIMA}_{\text{RoMa}}$. (i) GPDRNet. The red rectangles indicate the regions of global affine deformation, the yellow rectangles highlight the areas of local nonrigid deformation, and the orange rectangles mark the locations within the global deformation regions where the geometric structures of typical ground features are prone to change. (c1)–(i1), (c2)–(i2), and (c3)–(i3) are enlarged subimages from the red, yellow, and orange rectangles of (c)–(i), respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

estimation, while the limited feature representation capability of GLOH fails to handle the complex nonlinear radiometric differences between optical and SAR images. Furthermore, conventional optical flow estimation approaches inherently face constraints in terms of noise robustness and their ability to accommodate complex deformations.

Compared to OS-Flow, RIFT shows improved registration performance. Although RIFT struggles with local nonrigid deformations between multimodal images, it can effectively address the geometric and radiometric differences between multimodal images through sparse feature point matching and outlier rejection strategies, thereby achieving better registration performance. However, its performance remains suboptimal when compared to other learning-based dense registration approaches. Benefiting from the advanced deep optical flow architecture and the reconstructed deep feature extraction network, OSFlowNet demonstrates superior performance compared to OS-Flow and RIFT. ADRNet further enhances registration performance by integrating affine and deformable registration modules, showing outstanding results in both *CMR* and *EPE* metrics. Notably, RoMa achieves significant improvements over ADRNet. On the OSdataset, RoMa elevates $CMR_{T\leq 1px}$ by 2.89% and reduces *EPE* by 0.56 pixels. On the WHU-OPT-SAR dataset, RoMa achieves a 2.65% higher $CMR_{T\leq 1px}$ and a 0.97-pixel lower *EPE* compared to ADRNet. $\text{MINIMA}_{\text{RoMa}}$ extends these gains by training on a large-scale synthetic multimodal dataset. On OS dataset, it boosts $CMR_{T\leq 3px}$ to 73.56% and reduces *EPE* to 2.44 pixels. On WHU-OPT-SAR, $\text{MINIMA}_{\text{RoMa}}$ attains a $CMR_{T\leq 3px}$ of 76.07% with an *EPE* of 2.35 pixels, showcasing its superior cross-modal robustness.

While these methods exhibit individual strengths, our proposed GPDRNet significantly surpasses all counterparts on both the OSdataset and WHU-OPT-SAR datasets. Specifically, our method achieves the highest percentages at all *CMR* metrics and demonstrates the lowest *EPE* values, indicating substantial advantages in terms of registration accuracy and reliability.

The comprehensive results presented in Tables I and II clearly demonstrate the overall superiority of our registration method over existing dense registration approaches. Furthermore, the robustness of our method is well-established, enabled by its advanced FE, multisupervised loss, and postprocessing module. This comprehensive design ensures high-precision registration across diverse challenging scenarios, effectively handling both global affine and local nonrigid deformations.

*2) Qualitative Analysis:* To provide an intuitive comparison of the registration performance across different methods, we present the visualization of registration results on representative scenarios. As shown in Figs. 8 and 9, the selected scenes include various typical ground features such as urban, suburban, agricultural, and water bodies. The checkerboard visualization pattern clearly reveals the registration quality between fixed and moving images, particularly highlighting local alignment accuracy. It can be seen that, although the multimodal images globally conform to an affine transformation, significant nonrigid deformations still exist locally, imposing higher demands on the robustness and accuracy of the registration methods. By comparing the performance in global affine deformation, local nonrigid deformation, and the preservation of geometric structures of typical features within global deformation regions, we systematically evaluate the performance of each method.

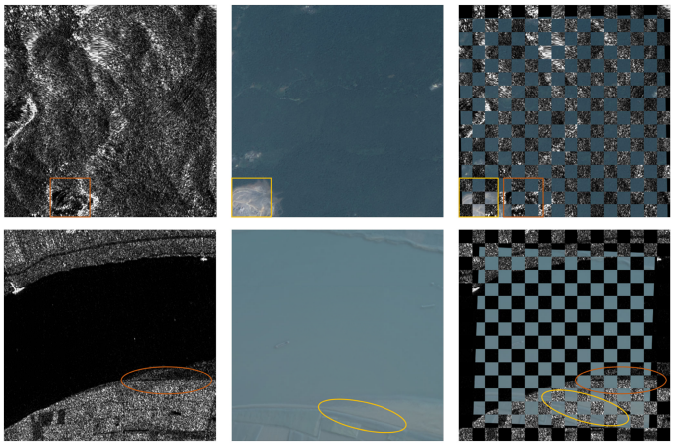
OS-Flow performs dense registration through optical flow estimation. Although theoretically capable of handling complex deformations, it completely fails in the presented scenario. It can be seen that OS-Flow struggles with scale and rotation differences between multimodal images, as well as local nonrigid deformations, often leading to distortion in the geometry of ground features. The fundamental limitation lies in its reliance on traditional handcrafted features with insufficient capability to model cross-modal complex deformations. The sparse feature matching mechanism of RIFT makes it more suitable for global registration but sensitive to local deformations. The registration results show that RIFT achieves coarse global alignment but still exhibits slight misalignments in detail and significant inaccuracies in local deformation regions.

OSFlowNet exhibits moderate registration performance overall, yet it shows insufficient robustness against significant scale and rotation differences, leading to noticeable alignment deviations under such transformations. While benefiting from a deep optical flow architecture that enhances its ability to model nonrigid deformations, OSFlowNet still struggles with alignment in highly distorted local areas. Furthermore, insufficient supervision constraints during training result in excessive freedom in optical flow field fitting, which fails to maintain the structural integrity of features in global deformation regions. In contrast, ADRNet demonstrates strong performance in preserving the geometry consistency of ground features by imposing multiple constraints on the dense warp field. However, limited by its feature representation capacity, ADRNet only achieves rough global alignment, and its fitting of local nonrigid deformation remains inadequate, failing to meet high-precision registration requirements.

RoMa employs anchor probability decoding to model multipeaked distributions and refines deformation fields with a fine feature pyramid, significantly improving dense deformation field fitting. Its enhanced version $\text{MINIMA}_{\text{RoMa}}$, trained on cross-modal synthetic data, further strengthens robustness against radiometric differences between multimodal images. Registration results demonstrate that both RoMa and $\text{MINIMA}_{\text{RoMa}}$ exhibit excellent performance in capturing both global affine and local nonrigid deformations. Nonetheless, influenced by mismatching noise existing in dense correspondences, they still fall short in preserving the geometric integrity of features across global deformation regions.

In comparison, our proposed GPDRNet demonstrates remarkable advantages in coordinated optimization of global and local deformations. The registered results show that GPDRNet delivers precise alignment in both global and local nonrigid deformation regions while effectively preserving the morphological integrity of typical features within global deformation areas. Qualitative results further confirm that, empowered by more discriminative feature representation, the specially designed GCL, and the postprocessing module, GPDRNet can accurately model both global affine

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 10. Representative failure cases of GPDRNet. (a) Fixed images with marked regions of interest. (b) Moving images with corresponding markers indicating roughly co-located areas. (c) Registration results, where misalignments reveal failures in low-texture scenarios.

TABLE III

ABLATION STUDY RESULTS OF THE PROPOSED METHODS

| Modules | | | Metrics | | | |
|---|---|---|---|---|---|---|
| FE | GCL | LDAPP | $CMR_{T\leq 1px}$ | $CMR_{T\leq 3px}$ | $CMR_{T\leq 5px}$ | *EPE* |
| | | | 22.96 | 70.10 | 88.84 | 2.99 |
| ✓ | | | 23.85 | 71.96 | 90.71 | 2.45 |
| | ✓ | | 24.47 | 71.04 | 90.13 | 2.51 |
| ✓ | ✓ | | 24.63 | 72.73 | 91.59 | 2.32 |
| ✓ | ✓ | ✓ | **25.98** | **74.85** | **92.33** | **2.26** |

and local nonrigid transformations simultaneously, enabling high-precision registration of multimodal images. Nevertheless, GPDRNet still faces certain challenges. As shown in Fig. 10, its registration performance degrades in low-texture regions, including large water bodies and dense vegetation. The lack of structural features or the presence of noise in these areas makes feature matching difficult, which in turn affects the accurate estimation of warp fields. These limitations are common to existing registration methods and point to meaningful directions for future research.

### D. Ablation Studies

In this section, we conduct ablation studies on OSdataset to evaluate the contributions of the FE module, the GCL, and the LDAPP module within our proposed GPDRNet.

The experimental results are presented in Table III. The first row shows the registration performance of the baseline model without any proposed modules, achieving a $CMR_{T\leq 1px}$ of 22.96%, $CMR_{T\leq 3px}$ of 70.10%, $CMR_{T\leq 5px}$ of 88.84%, and an *EPE* of 2.99 pixels. Compared to RoMa, our baseline, which incorporates the widely used $L_{\text{recon}}$ and $L_{\text{ssim}}$, already yields noticeable improvements across all metrics. This underscores the effectiveness of these auxiliary supervision terms in enhancing dense matching quality. The baseline model serves as a reference for assessing the improvements brought by each module. Introducing the FE module alone leads to noticeable gains across all metrics, indicating its capability to enhance feature representations and improve robustness against complex distortions. Similarly, incorporating the GCL refines the performance, underscoring the importance of enforcing geometry consistency for accurate image registration. The combination of both FE and GCL results in additional improvements, yielding gains of 1.67% in $CMR_{T\leq 1px}$, 2.63% in $CMR_{T\leq 3px}$, and 2.75% in $CMR_{T\leq 5px}$, while reducing the *EPE* by 0.67 pixels relative to the baseline. Finally, when all three modules are integrated, the model attains the best performance, achieving a $CMR_{T\leq 1px}$ of 25.98%, $CMR_{T\leq 3px}$ of 74.85%, $CMR_{T\leq 5px}$ of 92.33%, and an *EPE* of 2.26 pixels. These results highlight the synergistic effect of the FE module, the GCL, and the LDAPP module in our proposed method.

TABLE IV

EFFECTS OF APPENDING THE LDAPP MODULE TO DIFFERENT REGISTRATION NETWORKS ON THE OSDATASET

| Methods | LDAPP | Metrics | | | |
|---|---|---|---|---|---|
| | | $CMR_{T\leq 1px}$ | $CMR_{T\leq 3px}$ | $CMR_{T\leq 5px}$ | *EPE* |
| ADRNet | | 19.53 | 61.10 | 84.81 | 3.59 |
| ADRNet | ✓ | 18.78 | 60.09 | 84.55 | 3.74 |
| RoMa | | 22.42 | 69.27 | 87.86 | 3.03 |
| RoMa | ✓ | **23.56** | **70.04** | **88.90** | **2.92** |

TABLE V

CORRELATION BETWEEN PREDICTED CONFIDENCE AND REGISTRATION ACCURACY

| Confidence percentile | *EPE* | $CMR_{T\leq 1px}$ |
|---|---|---|
| Top 10 | **0.98** | **67.72** |
| Top 30 | 1.56 | 56.93 |
| Top 50 | 1.87 | 40.50 |

Moreover, we further investigate the generalizability of the LDAPP module. Specifically, we examine whether it can serve as a standalone postprocessing step to enhance the performance of other dense registration networks. To this end, we apply LDAPP directly to the warp fields output by RoMa and ADRNet. The results are summarized in Table IV.

When applied to the relatively clean and accurate warp field produced by RoMa, LDAPP improves registration performance across all evaluation metrics, confirming its capability to effectively decouple global and local deformations when provided with high-quality input. In contrast, applying LDAPP to the noisier output of ADRNet yields no improvement and even leads to slight degradation, demonstrating that its effectiveness is conditional and strongly depends on the quality of the initial warp field. These results underscore the importance of our integrated framework, in which the FE and the GCL work together to produce a refined and reliable initial warp field. This provides the essential foundation for LDAPP to successfully carry out deformation separation and refinement, thereby contributing to the overall performance gain of GPDRNet. In terms of computational cost, the LDAPP module introduces a modest runtime overhead. On an NVIDIA RTX 4090 GPU, GPDRNet without LDAPP processes a $512 \times 512$ image pair in 0.45 s on average, while the full pipeline requires 0.55 s. This represents a reasonable tradeoff given the consistent gains in registration accuracy across diverse deformation scenarios. To validate the effectiveness of the predicted confidence, we analyze the correlation between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
PERFORMANCE COMPARISON OF GPDRNET WITH DIFFERENT BACKBONES ON THE OSDATASET

| Backbones | | Metrics | | | |
|---|---|---|---|---|---|
| DINOv2 | DINOv3 | $CMR_{T\leq 1px}$ | $CMR_{T\leq 3px}$ | $CMR_{T\leq 5px}$ | *EPE* |
| ✓ | | 25.98 | 74.85 | 92.33 | 2.26 |
| | ✓ | **27.34** | **76.79** | **94.52** | **1.99** |

confidence percentiles and registration accuracy. As shown in Table V, pixels assigned higher confidence consistently exhibit lower *EPE* and higher $CMR_{T\leq 1px}$. For example, the top 10 percent highest-confidence pixels achieve an *EPE* of 0.98 pixels and a $CMR_{T\leq 1px}$ of 67.72%, which is significantly higher than the overall average. This ensures that the sparse points sampled for global affine estimation in LDAPP are reliable.

In addition to evaluating the contributions of the proposed modules, we further investigate the robustness and adaptability of our framework to more advanced pretrained feature extractors. DINOv3 [59], which has recently demonstrated superior representation capabilities, is adopted to replace the original DINOv2 backbone while keeping all other components equipped. The results in Table VI show that GPDRNet with DINOv3 achieves consistent and noticeable improvements across all evaluation metrics. This confirms that our method not only preserves its structural strengths but also effectively leverages enhanced feature representations to further boost registration accuracy.

## V. CONCLUSION

High-precision multimodal image registration is a key prerequisite for the integrated utilization and analysis of RS data. This study proposes a novel GPDRNet to address the challenge of coupled global affine transformations and local nonrigid deformations in multimodal RS images. GPDRNet enhances cross-modal semantic interaction through an FE module to improve feature discriminability and designs a GCL and an LDAPP module to preserve the structural integrity of typical ground features in global deformation regions. Comparative evaluations against state-of-the-art image registration methods on two public datasets validate the robustness and effectiveness of our approach.

In this work, model training and evaluation are conducted using synthetically generated datasets, as dense ground-truth warp fields are often unavailable for real complex deformation scenes. Additionally, GPDRNet remains challenged in low-texture regions. Future research will explore unsupervised dense registration frameworks to break through the bottleneck of labeled data dependency and enhance the robustness in low-texture regions, advancing the practical application of multimodal RS image dense registration.

## REFERENCES


[1] J. Qin, M. Li, G. Armin, J. Gong, J. Zhong, and X. Liao, "Efficient coral survey using aerial remote sensing and multi-modal segmentation for large-scale ecological assessment," *Photogrammetric Eng. Remote Sens.*, vol. 91, no. 8, pp. 509–516, Aug. 2025.

[2] Y. Ye et al., "Tuple perturbation-based contrastive learning framework for multimodal remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5613615.

[3] M. Sharma et al., "YOLOrs: Object detection in multimodal remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1497–1508, 2021.

[4] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415.

[5] S. Chirakkal, F. Bovolo, A. R. Misra, L. Bruzzone, and A. Bhattacharya, "A general framework for change detection using multimodal remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10665–10680, 2021.

[6] Z. Lv, H. Huang, W. Sun, T. Lei, J. A. Benediktsson, and J. Li, "Novel enhanced UNet for change detection using multimodal remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[7] X. Jiang, J. Ma, G. Xiao, Z. Shao, and X. Guo, "A review of multimodal image matching: Methods and applications," *Inf. Fusion*, vol. 73, pp. 22–71, Sep. 2021.

[8] N. Merkle, S. Auer, R. Müller, and P. Reinartz, "Exploring the potential of conditional adversarial networks for optical and SAR image matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 1811–1820, Jun. 2018.

[9] Y. Ye, L. Shen, M. Hao, J. Wang, and Z. Xu, "Robust optical-to-SAR image matching based on shape properties," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 564–568, Apr. 2017.

[10] Y. Yao, Y. Zhang, Y. Wan, X. Liu, X. Yan, and J. Li, "Multi-modal remote sensing image matching considering co-occurrence filter," *IEEE Trans. Image Process.*, vol. 31, pp. 2584–2597, 2022.

[11] Y. Ye, B. Zhu, T. Tang, C. Yang, Q. Xu, and G. Zhang, "A robust multimodal remote sensing image registration method and system using steerable filters with first- and second-order gradients," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 331–350, Jun. 2022.

[12] H. Zhang et al., "Registration of multimodal remote sensing image based on deep fully convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3028–3042, Aug. 2019.

[13] Y. Ye, T. Tang, B. Zhu, C. Yang, B. Li, and S. Hao, "A multiscale framework with unsupervised learning for remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622215.

[14] C. Yang, Y. Ye, R. Liu, C. Fan, L. Zhou, and J. Deng, "Suitable area selection method based on scene matching level segmentation," *Chin. J. Aeronaut.*, vol. 39, no. 2, 2025, Art. no. 103719.

[15] A. Wong and D. A. Clausi, "ARRSI: Automatic registration of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 5, pp. 1483–1493, May 2007.

[16] Z. Song, S. Zhou, and J. Guan, "A novel image registration algorithm for remote sensing under affine transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4895–4912, Aug. 2014.

[17] F. Cigna, L. B. Bateson, C. J. Jordan, and C. Dashwood, "Simulating SAR geometric distortions and predicting persistent scatterer densities for ERS-1/2 and ENVISAT C-band SAR and InSAR applications: Nationwide feasibility assessment to monitor the landmass of great Britain with SAR imagery," *Remote Sens. Environ.*, vol. 152, pp. 441–466, Sep. 2014.

[18] J. Ma, H. Zhou, J. Zhao, Y. Gao, J. Jiang, and J. Tian, "Robust feature matching for remote sensing image registration via locally linear transforming," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6469–6481, Dec. 2015.

[19] L. Yang, X. Li, J. Yu, L. Jiang, X. Su, and F. Chen, "Thin-plate-spline-based registration for correcting local nonlinear deformation of bidirectional whisk-broom images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13332–13345, 2024.

[20] H. Zhang et al., "Optical and SAR image dense registration using a robust deep optical flow framework," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1269–1294, 2023.

[21] W. Wang, K. Mu, and H. Liu, "A multihierarchy flow field prediction network for multimodal remote sensing image registration," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5232–5243, 2025.

[22] Y. Xiao, C. Zhang, Y. Chen, B. Jiang, and J. Tang, "ADRNet: Affine and deformable registration networks for multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5207613.

[23] B. Zhu, L. Zhou, S. Pu, J. Fan, and Y. Ye, "Advances and challenges in multimodal remote sensing image registration," *IEEE J. Miniaturization Air Space Syst.*, vol. 4, no. 2, pp. 165–174, Jun. 2023.

[24] Y. Ye, L. Bruzzone, J. Shan, F. Bovolo, and Q. Zhu, "Fast and robust matching for multimodal remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9059–9070, Nov. 2019.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.

[25] J. N. Sarvaiya, S. Patnaik, and S. Bombaywala, "Image registration by template matching using normalized cross-correlation," in *Proc. Int. Conf. Adv. Comput., Control, Telecommun. Technol.*, Dec. 2009, pp. 819–822.

[26] Y. Ye, J. Shan, L. Bruzzone, and L. Shen, "Robust registration of multimodal remote sensing images based on structural similarity," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2941–2958, May 2017.

[27] J. Fan, Y. Wu, M. Li, W. Liang, and Y. Cao, "SAR and optical image registration using nonlinear diffusion and phase congruency structural descriptor," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5368–5379, Sep. 2018.

[28] X. Xiong, Q. Xu, G. Jin, H. Zhang, and X. Gao, "Rank-based local self-similarity descriptor for optical-to-SAR image matching," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1742–1746, Oct. 2020.

[29] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[30] H. Bay, T. Tuytelaars, and L. V. Gool, "SURF: Speeded up robust features," in *Proc. ECCV*, 2006, pp. 404–417.

[31] J. Li, Q. Hu, and M. Ai, "RIFT: Multi-modal image matching based on radiation-variation insensitive feature transform," *IEEE Trans. Image Process.*, vol. 29, pp. 3296–3310, 2020.

[32] L. Li, L. Han, Y. Ye, Y. Xiang, and T. Zhang, "Deep learning in remote sensing image matching: A survey," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 88–112, Jul. 2025.

[33] N. Merkle, W. Luo, S. Auer, R. Müller, and R. Urtasun, "Exploiting deep matching and SAR data for the geo-localization accuracy improvement of optical satellite images," *Remote Sens.*, vol. 9, no. 6, p. 586, Jun. 2017.

[34] H. Zhang et al., "Explore better network framework for high-resolution optical and SAR image matching," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4704418.

[35] L. H. Hughes, D. Marcos, S. Lobry, D. Tuia, and M. Schmitt, "A deep learning framework for matching of SAR and optical imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 166–179, Nov. 2020.

[36] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 224–236.

[37] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4938–4947.

[38] P. Lindenberger, P. E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17627–17638.

[39] Z. Chen, X. Zhang, X. Xu, H. Chen, X. Mi, and J. Yang, "Registration-aware cross-modal interaction network for optical and SAR images," *Inf. Fusion*, vol. 123, Nov. 2025, Art. no. 103278.

[40] B. D. Lucas and T. Kanade, "An iterative image registration technique with an application to stereo vision," in *Proc. 7th Int. Joint Conf. Arti. Intell. (IJCAI)*, 1981, pp. 674–679.

[41] A. Plyer, E. Colin-Koeniguer, and F. Weissgerber, "A new coregistration algorithm for recent applications on urban SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2198–2202, Nov. 2015.

[42] G. Brigot, E. Colin-Koeniguer, A. Plyer, and F. Janez, "Adaptation and evaluation of an optical flow method applied to coregistration of forest remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 7, pp. 2923–2939, Jul. 2016.

[43] Y. Xiang, F. Wang, L. Wan, N. Jiao, and H. You, "OS-flow: A robust algorithm for dense optical and SAR image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6335–6354, Sep. 2019.

[44] B. K. P. Horn and B. G. Schunck, "Determining optical flow," *Artif. Intell.*, vol. 17, nos. 1–3, pp. 185–203, Aug. 1981.

[45] Q. Yu, Y. Jiang, W. Zhao, and T. Sun, "High-precision pixelwise SAR–optical image registration via flow fusion estimation based on an attention mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3958–3971, 2022.

[46] D. Ibañez, R. Fernandez-Beltran, and F. Pla, "FloU-Net: An optical flow network for multimodal self-supervised image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[47] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[48] Z. Teed and J. Deng, "RAFT: Recurrent all-pairs field transforms for optical flow," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 402–419.

[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[50] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[51] J. Edstedt, I. Athanasiadis, M. Wadenbäck, and M. Felsberg, "DKM: Dense kernelized feature matching for geometry estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 17765–17775.

[52] J. Edstedt, Q. Sun, G. Bökman, M. Wadenbäck, and M. Felsberg, "RoMa: Robust dense feature matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 19790–19800.

[53] J. Ren, X. Jiang, Z. Li, D. Liang, X. Zhou, and X. Bai, "MINIMA: Modality invariant image matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 23059–23068.

[54] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[55] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[56] L. Nie, C. Lin, K. Liao, S. Liu, and Y. Zhao, "Parallax-tolerant unsupervised deep image stitching," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 7399–7408.

[57] Y. Xiang, R. Tao, F. Wang, H. You, and B. Han, "Automatic registration of optical and SAR images via improved phase congruency model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5847–5861, 2020.

[58] X. Li et al., "MCANet: A joint semantic segmentation framework of optical and SAR images for land use classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102638.

[59] O. Siméoni et al., "DINOv3," 2025, *arXiv:2508.10104*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:40:08 UTC from IEEE Xplore. Restrictions apply.